

ARVIND KUMAR VERMA

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AI/ML Engineer with hands-on experience building and deploying machine learning, computer vision, and full-stack software applications. Developed an end-to-end waste image classification system achieving ~91% test accuracy using PyTorch, ResNet, Docker, and Streamlit.

WORK EXPERIENCE

AI Research Intern (Software Development)

Jul 2026 – Sep 2026 (12 weeks)

Space Applications Centre (SAC), ISRO, Ahmedabad, Gujarat

- Independently designed and built [OC-ECV Local Engine](#), a local-first desktop application (Tauri, React/TypeScript, Python/FastAPI) for ingesting, processing, and visualizing Ocean Color and Essential Climate Variable satellite data, delivered as a fully packaged, installable application within a strict deadline.
- Validated the system end-to-end against real satellite data from 6 missions (MODIS-Aqua, SMAP, CCMP, SWOT, Oceansat-3) across 11 of 13 target climate-variable categories; diagnosed a performance bottleneck as I/O-bound rather than CPU-bound and implemented a local caching layer achieving a ~55x speedup on repeated queries.
- Continuing development post-internship to implement a planned ML-based analysis module (e.g., automated algal-bloom detection) as part of the project's scoped future roadmap, applying deep learning and computer vision techniques to real-world satellite geospatial data.

AI Developer Intern

Jan 2026 – Apr 2026 (14 weeks)

Cognyx AI Pvt. Ltd.

- Developed an end-to-end AI-based Waste Image Classification System using deep learning and computer vision to automatically classify waste materials (glass, metal, paper, plastic) for efficient waste segregation.
- Implemented transfer learning with ResNet architectures, applying fine-tuning and differential learning rates to achieve ~91% test accuracy on the final model.
- Built and managed the complete ML pipeline — data preprocessing, model training, evaluation, experiment tracking, and deployment.
- Deployed a Streamlit web application enabling real-time image upload and prediction with confidence scores, containerised using Docker.

Data Analyst Intern

Jul 2025 (2 weeks)

IBM CSRBOX – Summer GTU Internship Program

- Performed data preprocessing on agricultural datasets including handling missing values, outlier removal, and data cleaning using Microsoft Excel.
- Designed and built an interactive dynamic Tableau dashboard (AgriPulse) using agricultural data to help farmers make informed, data-driven decisions.

PROJECTS

[Waste Image Classifier](#) — (PyTorch, ResNet, Streamlit, Docker)

[Crop Disease Classification](#) — (Jupyter Notebook, PyTorch, Scikit-learn)

- Balanced an imbalanced 5-class cassava disease image dataset (healthy, mosaic disease, brown streak disease, green mottle, bacterial blight) via normalization and undersampling, then built a custom CNN (3 convolution + max-pooling layers, followed by flattening and 2 dense layers) for classification.
- Diagnosed overfitting in the baseline CNN and resolved it by implementing transfer learning with ResNet-50 alongside learning-rate scheduling, checkpointing, and early-stopping callbacks, achieving good classification results on the held-out test set.

[Wildlife Conservation in Côte d'Ivoire](#) — (Jupyter Notebook, PyTorch, Scikit-learn)

- Built a CNN-based multiclass classifier to categorize camera-trap animal images into 8 species classes (antelope-duiker, bird, blank, civet-genet, hog, leopard, monkey-prosimian, rodent), achieving good classification results to support wildlife-monitoring efforts.

[Insurance Premium Category Prediction](#) — (FastAPI, Docker, Scikit-learn, Jupyter Notebook, Streamlit)

- Built a Random Forest Classifier to predict insurance premium category (High / Medium / Low), achieving 94% accuracy on the test set, served through a FastAPI backend containerized with Docker and exposed via an interactive Streamlit frontend.

AgriPulse: Visualizing Profitability and Sustainability in Agriculture — (Microsoft Excel, Tableau) | [Live Dashboard](#)

SKILLS

Programming: Python (Advanced), Java, SQL

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Hyperparameter Tuning, CNNs, Transfer Learning

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn

Computer Vision: Image Classification, Object Detection, Semantic Segmentation, Data/Bounding Box/Keypoint Annotation, Single Object Tracking

Software Engineering: Full-stack desktop apps (Tauri, React/TypeScript, FastAPI), geospatial data processing (xarray, GDAL, rasterio)

Tools & OS: Docker, Git, Streamlit, FastAPI, CVAT, Tableau, Linux (Ubuntu), Windows

Libraries: NumPy, Pandas, Matplotlib, Seaborn

EDUCATION

New L.J. Institute of Engineering & Technology, Ahmedabad

2022 – 2026

Bachelor of Engineering (AI/ML) – CPI: 9.54 | CGPA: 9.52 | Languages: English, Hindi